

Dr. Nitin Lohan

National Post-Doctoral Fellow (NPDF) · Atmospheric Scientist

Department of Earth and Atmospheric Sciences, National Institute of Technology Rourkela,
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RESEARCH PROFILE

Applied mathematician and atmospheric scientist, currently a National Post-Doctoral Fellow at NIT Rourkela working on numerical and machine-learning forecasting over the Himalayan region. Research covers the diagnosis, simulation and statistics of extreme weather over India: Himalayan cloudbursts, monsoon low-pressure systems, tropical cyclones, heatwaves and the changing spatial structure of monsoon rainfall extremes. The work combines WRF/WRFDA modeling, 3DVAR data assimilation, extreme-value statistics and reanalysis and satellite datasets (ERA5, IMDAA, IMD, GPM, INSAT), in collaboration with scientists at IMD, NCMRWF and IITM.

8

Peer-reviewed journal articles

4

Conference papers & book chapters

3

Conference presentations

NET ×2 · GATE

National qualifications

RESEARCH INTERESTS

- Numerical and machine-learning forecasting over the Himalayan region
- Extreme weather: cloudbursts, monsoon low-pressure systems, tropical cyclones and heatwaves
- Numerical weather prediction and data assimilation (WRF, WRFDA, 3DVAR)
- Extreme-value statistics and spatial dependence of monsoon rainfall extremes
- Climate change, urbanization and associated socio-economic impacts
- Big-data analysis of reanalysis and satellite datasets; remote sensing and GIS

EDUCATION

Ph.D. in Applied Mathematics

2020 – 2026

Gautam Buddha University, Greater Noida, India

Thesis: *Diagnostic and Numerical Modeling Study of Heavy Precipitation Events over Indian Region*

Supervisor: Dr. Sushil Kumar · Defended 31 January 2026 · CGPA 8.29

M.Sc. in Mathematics

2016 – 2018

DAV College, Panjab University, Chandigarh, India

B.Sc. (Physics, Chemistry, Mathematics)

2013 – 2016

Government PG College Jind, Kurukshetra University, Haryana, India

ACADEMIC & PROFESSIONAL EXPERIENCE

National Post-Doctoral Fellow (NPDF)

Oct 2026 – Present

Department of Earth and Atmospheric Sciences, National Institute of Technology Rourkela, Odisha

- Research on numerical and machine-learning forecasting over the Himalayan region.

Assistant Professor (Mathematics)

Jul 2025 – Sep 2026

G.L. Bajaj Institute of Technology and Management, Greater Noida

- Taught undergraduate engineering mathematics to 100+ students; developed course material, assessments and problem sets.
- Mentored students in applied mathematics, numerical methods and scientific computing with Python.

- Continued research on atmospheric modeling, extreme-precipitation diagnostics and rainfall extremes alongside teaching.

AI Trainer / Evaluator & STEM Benchmark Developer (Remote)

Sep 2025 – Present

Mercor · Turing · micro1 · Handshake AI

- Design high-difficulty mathematics, numerical-methods and inverse-problem tasks, SciCode-style scientific-coding benchmarks and RLHF preference data for training and evaluating large language models.

Doctoral Researcher (Applied Mathematics)

Jul 2020 – Jan 2026

Gautam Buddha University, Greater Noida

- Diagnostic and numerical-modeling studies of heavy-precipitation and cloudburst events over the Indian region.
- Configured and ran WRF/WRFDA simulations on HPC clusters; applied 3DVAR data assimilation to high-impact case studies.
- Analyzed ERA5, IMDAA, IMD, GPM and INSAT datasets for extreme-rainfall and monsoon-system diagnostics.

PUBLICATIONS

Journal Articles

- [1] **Lohan, N.**, Kumar, S., Ben Hasher, F. F., & Zhran, M. (2026). Reorganization of the spatial dependence structure of Indian summer monsoon rainfall extremes, 1951–2025. *Atmosphere*, 17(9), 892. doi:10.3390/atmos17090892
- [2] **Lohan, N.**, Pattanaik, D. R., Kumar, S., & Giri, R. K. (2025). Diagnostic study of monsoon-linked rainfall events: Unraveling the role of monsoon low pressure systems across the Indian subcontinent. *Earth Systems and Environment*, 1–14. doi:10.1007/s41748-025-00706-y
- [3] **Lohan, N.**, Routray, A., Kumar, R., Mahala, B. K., & Kumar, S. (2025). Validation and diagnostic study of cloudburst events over the Himalayan region using IMDAA and ERA5. *Geomatics, Natural Hazards and Risk*, 16(1), 2446589. doi:10.1080/19475705.2024.2446589
- [4] **Lohan, N.**, Kumar, S., Singh, V., Gupta, R. P., & Tiwari, G. (2025). Analyzing an extreme rainfall event in Himachal Pradesh, India, to contribute to sustainable development. *Sustainability*, 17(5), 2115. doi:10.3390/su17052115
- [5] Singh, N., Vinayak Shukla, M., **Lohan, N.**, Mitra, A. K., & Giri, R. K. (2025). Evaluation of INSAT-3DR sounder derived wind index using radiosonde observation over India. *Theoretical and Applied Climatology*, 156(1), 58. doi:10.1007/s00704-024-05284-z
- [6] Kumar, A., Kumar, S., Kumar, N., **Lohan, N.**, & Nupur, R. (2025). Estimating the influence of cumulus parametrization schemes in simulation of super cyclonic storm ‘Amphan’ over the Bay of Bengal using WRF-ARW model. *Indian Journal of Pure & Applied Physics*, 63(10). doi:10.56042/ijpap.v63i10.11386
- [7] Kumar, A., Kumar, S., Kumar, N., **Lohan, N.**, & Bhatla, R. (2024). Comparison of characteristics associated with super cyclonic storm ‘Amphan’ using numerical model WRF-ARW analysis and ERA5 reanalysis. *Journal of the Indian Geophysical Union*, 28(2), 99–107.
- [8] **Lohan, N.**, Kumar, S., Mishra, P., Kumar, S., & Giri, R. K. (2023). A study of the long-term trend of methane emission over the Indian region during 1970–2020. *Vayu Mandal*, 49(2), 26–33. Indian Meteorological Society.

Conference Proceedings & Book Chapters

- [9] Kumari, N., Singh, N., Panda, B., Singh, V., Kumar, N., & **Lohan, N.** (2026). Simulation of heatwave event using WRF model across Uttar Pradesh, India. In *2026 2nd International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI)*. IEEE. doi:10.1109/IC3ECSBHI67834.2026.11468976

- [10] Panda, J., Paul, D., Sarkar, A., Mukherjee, A., Bhasi, I., Tom, G., **Lohan, N.**, & Kumar, S. (2025). Severe storms in changing climate scenario with a distinctive emphasis on South Asia and India. In *Severe Storms: Anatomy, Early Warning Systems and Aftermath in Changing Climate Scenarios*, pp. 563–606. Springer Nature Singapore. **BOOK CHAPTER** doi:10.1007/978-981-97-7075-5_22
- [11] Kumar, N., Kumar, S., Panda, B., Routray, A., Kumari, N., & **Lohan, N.** (2025). Diagnostic study of thunderstorms over Uttar Pradesh, India. In *2025 International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI)*, pp. 1276–1279. IEEE. doi:10.1109/IC3ECSBHI63591.2025.10990776
- [12] **Lohan, N.**, Tiwari, G., Kumar, S., & Routray, A. (2023). On the analysis of a very severe cyclonic storm Nilofar over the Arabian Sea: A numerical weather prediction model study. In *International Conference on Applied and Industrial Mathematics (ICAIM 2023)*, pp. 157–169. Springer Nature Singapore. doi:10.1007/978-981-96-3460-6_10

CONFERENCE PRESENTATIONS

- Role of Monsoon Low Pressure Systems in Extreme Rainfall Events over the Indian Subcontinent** 2023
Oral presentation · TROPMET 2023, Jaipur Chapter, Birla Institute of Scientific Research, Jaipur, India
N. Lohan, S. Kumar, R. K. Giri and D. R. Pattanaik
- On the Analysis of a Very Severe Cyclonic Storm Nilofar over the Arabian Sea: A Numerical Weather Prediction Model Study** 2023
Oral presentation · ICAIM 2023, Sharda University, Greater Noida, India
N. Lohan, S. Kumar and G. Tiwari
- Diagnostic Study of Cloudburst Events over the Himalayan Region using Reanalysis Datasets** 2022
Poster presentation · TROPMET 2022, Bhopal Chapter, IISER Bhopal, India
N. Lohan, S. Kumar and G. Tiwari

RESEARCH COLLABORATIONS

- **National Centre for Medium Range Weather Forecasting (NCMRWF)**: collaborative research with Dr. Ashish Routray (Scientist-F); hands-on experience in model execution.
- **India Meteorological Department (IMD)**: HPC-based modeling experiments with Dr. R. K. Giri (Scientist-F) and Dr. D. R. Pattanaik (Scientist-G); exposure to IMD operational workflows and 3DVAR data assimilation.
- **Indian Institute of Tropical Meteorology (IITM), Pune**: joint work on WRF-based extreme-rainfall studies with Dr. Vivek Singh (Scientist-D).

QUALIFICATIONS, WORKSHOPS & TRAINING

- CSIR-UGC National Eligibility Test (NET), Mathematical Sciences** 2019, 2022
GATE, Mathematics 2022
- Induction Training Program on Meteorology (online)** Apr – May 2024
Meteorological Training Institute, India Meteorological Department, Pune
- Regional Capacity Building Programme (Series 3): Inspiring the Minds for Disaster Risk Reduction** Feb 2023
National Institute of Disaster Management (NIDM), India. Five-day programme
- National Workshop on Leadership for Promoting Ethics and Morality in Research** Mar 2020
Panjab University, Chandigarh. Two-day workshop

TECHNICAL SKILLS

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|--------------------|---|
| Programming | Python (expert): NumPy, SciPy, pandas, xarray, Matplotlib, Cartopy, wrf-python; machine learning (intermediate) |
| Statistics | Extreme-value statistics (extremal coefficient, F-madogram, Brown–Resnick dependence models); trend and change-point analysis (Pettitt test); error metrics (MAE, RMSE, correlation) for model verification |

Modeling & HPC	WRF-ARW, WRFDA (3DVAR), WRF installation; HPC job handling with SLURM/PBS; parallel computing; automation and scheduling
Data & Visualization	NetCDF/HDF data, CDO, GrADS, NCView, Panoply, HDFView, ArcGIS; ERA5, IMDAA, IMD, GPM and INSAT datasets
Writing & Tools	LaTeX, Git/GitHub, JupyterLab, Linux
Languages	Hindi (native), English (fluent)

REFERENCES

Dr. Sushil Kumar

Former Head & Assistant
Professor
Gautam Buddha University,
Greater Noida
(Ph.D. supervisor)

Dr. R. K. Giri

Scientist-F
India Meteorological Department
New Delhi

Dr. Vivek Singh

Scientist-D
Indian Institute of Tropical
Meteorology
Pune

Contact details of referees available on request.